

HDOC DAY 42

PROBLEM

Write a menu-driven C program to perform the following operations using separate functions:

- Remove all vowels from a string.
- Find the first repeating lowercase alphabet in a string.
- Check if two strings are anagrams of each other.

CODE

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

// Function to remove all vowels from a string
void removeVowels(char str[])
{
    int i, j = 0;
    for (i = 0; str[i] != '\0'; i++)
    {
        char ch = tolower((unsigned char)str[i]);

        if (ch != 'a' && ch != 'e' && ch != 'i' &&
            ch != 'o' && ch != 'u')
        {
            str[j++] = str[i];
        }
    }
}
```

```

    }

    str[j] = '\0';

    printf("String after removing vowels: %s\n", str);
}

// Function to find the first repeating lowercase alphabet
void firstRepeating(char str[])
{
    int count[26] = {0};
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            count[str[i] - 'a']++;

            if (count[str[i] - 'a'] == 2)
            {
                printf("First repeating lowercase alphabet: %c\n",
                    str[i]);
                return;
            }
        }
    }

    printf("No repeating lowercase alphabet found.\n");
}

```

```
}
```

```
// Function to check whether two strings are anagrams
```

```
void checkAnagram(char str1[], char str2[])
```

```
{
```

```
    int count[256] = {0};
```

```
    int i;
```

```
    for (i = 0; str1[i] != '\0'; i++)
```

```
    {
```

```
        unsigned char ch = (unsigned char)str1[i];
```

```
        if (!isspace(ch))
```

```
            count[tolower(ch)]++;
```

```
    }
```

```
    for (i = 0; str2[i] != '\0'; i++)
```

```
    {
```

```
        unsigned char ch = (unsigned char)str2[i];
```

```
        if (!isspace(ch))
```

```
            count[tolower(ch)]--;
```

```
    }
```

```
    for (i = 0; i < 256; i++)
```

```
    {
```

```
        if (count[i] != 0)
```

```
        {
```

```

        printf("The strings are not anagrams.\n");
        return;
    }
}

printf("The strings are anagrams.\n");
}

int main()
{
    char str1[200], str2[200];
    int choice;

    do
    {
        printf("\n----- MENU ----- \n");
        printf("1. Remove vowels from a string\n");
        printf("2. Find the first repeating lowercase alphabet\n");
        printf("3. Check whether two strings are anagrams\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar(); // Consume newline

        switch (choice)
        {
            case 1:
                printf("Enter a string: ");

```

```
fgets(str1, sizeof(str1), stdin);  
str1[strcspn(str1, "\n")] = '\0';
```

```
removeVowels(str1);  
break;
```

case 2:

```
printf("Enter a string: ");  
fgets(str1, sizeof(str1), stdin);  
str1[strcspn(str1, "\n")] = '\0';
```

```
firstRepeating(str1);  
break;
```

case 3:

```
printf("Enter the first string: ");  
fgets(str1, sizeof(str1), stdin);  
str1[strcspn(str1, "\n")] = '\0';
```

```
printf("Enter the second string: ");  
fgets(str2, sizeof(str2), stdin);  
str2[strcspn(str2, "\n")] = '\0';
```

```
checkAnagram(str1, str2);  
break;
```

case 4:

```
printf("Exiting program.\n");
```

```
break;
```

```
default:
```

```
    printf("Invalid choice! Please try again.\n");
```

```
}
```

```
} while (choice != 4);
```

```
return 0;
```

```
}
```